A Proof of Butler's Positivity Conjecture

arXiv:2608.11543

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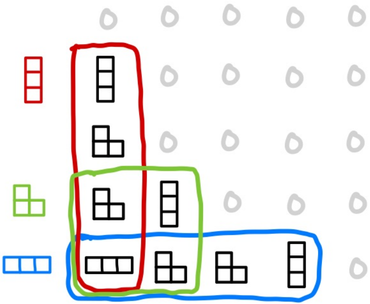
Section 1

Butler's Positivity Conjecture



Macdonald Polynomials

Schur [1901] $s_\lambda(X) \in \Lambda$ Schur function

$$\Downarrow$$

Littlewood [1961] $P_\lambda(X;t) \in \Lambda_t$ Hall–Littlewood symmetric function

$$\Downarrow$$

Macdonald [1988] $P_\lambda(X;q,t) \in \Lambda_{q,t}$ Macdonald symmetric function

There are different forms of Macdonald functions

$$P_\lambda = m_\lambda + (\text{lower term})$$

$$J_\lambda = P_\lambda \cdot \prod_{\square\in\lambda}(1 - q^{a(\square)} t^{\ell(\square)+1})$$

$$H_\lambda = J_\lambda[\tfrac{X}{1-t}] = J_\lambda \begin{pmatrix} x_1 & tx_1 & t^2x_1 & \dots \\ x_2 & tx_2 & t^2x_2 & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix}$$

$$\widetilde{H}_\lambda = t^{n(\lambda)} H_\lambda|_{t\mapsto t^{-1}}$$

Theorem (Haiman, 2001)

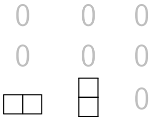
$$\widetilde{H}_\lambda \in \sum \mathbb{N}[q,t]\cdot s_\theta.$$

This was conjectured by Macdonald.

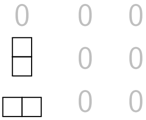
Example

$$\widetilde{H}_{(1)} = s_{(1)}$$

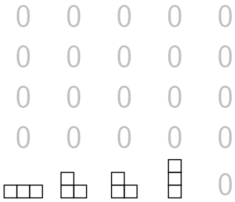
$$\widetilde{H}_{(2)} = s_{(2)} + qs_{(1,1)}$$



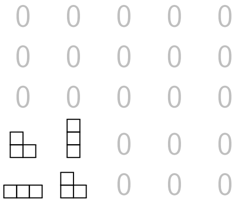
$$\widetilde{H}_{(2)} = s_{(2)} + ts_{(1,1)}$$



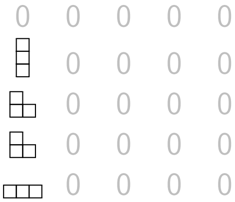
$$\widetilde{H}_{(3)}$$



$$\widetilde{H}_{(2,1)}$$



$$\widetilde{H}_{(1,1,1)}$$



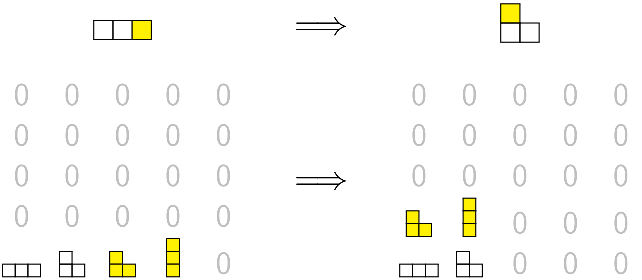
Butler's conjecture

Observing from the computation, Butler conjectured that

if μ is obtained by moving a box of λ, then

$\widetilde{H}_\mu$ is obtained by moving a summand of $\widetilde{H}_\lambda$.

For example,



Butler's conjecture

Note that if $\mu \vdash n$,

$$\widetilde{H}_\mu = 1 \cdot h_n + \cdots + T_\mu \cdot e_n,$$

where $T_\mu = \prod_{(i,j)\in D(\mu)} q^i t^j = q^{n(\mu')} t^{n(\mu)}$. The precise conjecture can be stated as follows. we can write

$$\widetilde{H}_\lambda(X;q,t) = 1 \cdot I_{\lambda,\mu}(X;q,t) + T_\lambda \cdot J_{\lambda,\mu}(X;q,t)$$
$$\widetilde{H}_\mu(X;q,t) = 1 \cdot I_{\lambda,\mu}(X;q,t) + T_\mu \cdot J_{\lambda,\mu}(X;q,t),$$

such that

$$I_{\lambda,\mu}(X;q,t),\ J_{\lambda,\mu}(X;q,t) \in \sum \mathbb{N}[q,t] \cdot s_\theta(X).$$

Butler's conjecture

Since $T_\lambda \neq T_\mu$, we have

$$I_{\lambda,\mu}(X;q,t) = \frac{T_\lambda \widetilde{H}_\mu(X;q,t) - T_\mu \widetilde{H}_\lambda(X;q,t)}{T_\lambda - T_\mu}. \qquad (*)$$

Using the reciprocity identity:

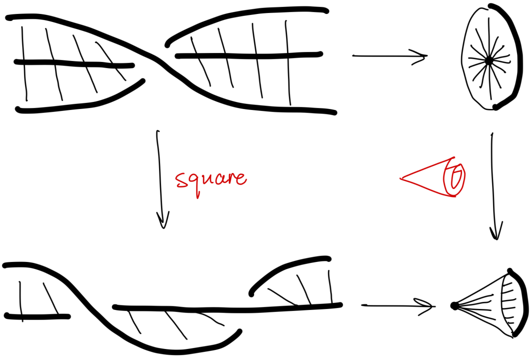
$$\widetilde{H}_\mu(X;q,t) = t^{n(\mu)} q^{n(\mu')} \omega \widetilde{H}_\mu(X;q^{-1},t^{-1}),$$

it is easy to see $J_{\lambda,\mu}(X;q,t) = \omega I_{\lambda,\mu}(X;q^{-1},t^{-1})$. So the conjecture is equivalent to

$$(*) \in \sum \mathbb{N}[q,t] \cdot s_\theta(X).$$

Section 2

Hilbert Schemes



Geometry of Hilbert schemes

Consider the **Hilbert scheme**

$$\begin{aligned} H_n &= \text{Hilbert scheme of } n\text{-points in } \mathbb{C}^2 \\ &= \{ I \trianglelefteq \mathbb{C}[x,y] : \dim_{\mathbb{C}} \mathbb{C}[x,y]/I = n \}. \end{aligned}$$

By Fogarty, H_n is a nonsingular variety of dimension $2n$.

Also consider the **symmetric power**

$$\begin{aligned} S^n\mathbb{C}^2 &= (\mathbb{C}^2)^n/S_n = \operatorname{Spec} \mathbb{C}[x_1, \ldots, x_n, y_1, \ldots, y_n]^{S_n} \\ &= \{\text{muti-sets of } n\text{-points in } \mathbb{C}^2\}. \end{aligned}$$

We have the **Hilbert–Chow** map $\sigma : H_n \longrightarrow S^n\mathbb{C}^2$ by sending I to the support (with multiplicity) of $\mathbb{C}[x,y]/I$.

Geometry of Hilbert schemes

The **isospectral Hilbert scheme** X_n is defined to be the reduced fiber product in the following diagram

$$\begin{tikzcd} X_n \arrow[r] \arrow[d, "\rho"'] & \mathbb{C}^{2n} \arrow[d] \\ H_n \arrow[r, "\sigma"] & S^n\mathbb{C}^{2n}. \end{tikzcd}$$

Theorem (Haiman)

The map ρ is finite and flat.

The **Procesi bundle** is the sheaf

$$\mathcal{P} = \rho_* \mathcal{O}_{X_n}.$$

This is a vector bundle of rank $n!$ by Haiman's Theorem!

Example ($n = 1$)

When $n = 1$, all spaces are $\mathbb{C}^2$, and all morphisms are identity.

$$\begin{tikzcd}
\mathbb{C}^2 = X_n \arrow[r] \arrow[d, "\rho"'] & \mathbb{C}^{2n} = \mathbb{C}^2 \arrow[d] \\
\mathbb{C}^2 = H_n \arrow[r, "\sigma"] & S^n\mathbb{C}^2 = \mathbb{C}^2
\end{tikzcd}$$

Any $I \in H_n$ is a maximal ideal, i.e. it takes the form $\langle x - x_0, y - y_0 \rangle$ for some $(x_0, y_0) \in \mathbb{C}^2$. So $H_n = \mathbb{C}^2$.

Since S_n is trivial, all others are also $\mathbb{C}^2$.

Example ($n = 2$)

When $n = 2$, we can identify H_n with the variety of

- two distinct points in $\mathbb{C}^2$ (**unordered**);
- a point with a tangent direction in $\mathbb{C}^2$.

We can identify $H_n = \mathbb{C}^2 \times$ total space of $\mathcal{O}_{\mathbb{P}^1}(-2)$.

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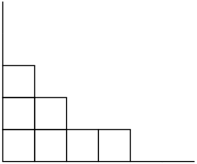
We can identify $X_n = \mathbb{C}^2 \times$ total space of $\mathcal{O}_{\mathbb{P}^1}(-1)$.

The map $\rho : X_n \to H_n$ is by taking square, which is flat.

Geometry of Hilbert schemes

For each partition $\mu \vdash n$, we can define a monomial ideal $I_\mu \in H_n$ as follows.



$\mu = (4, 3, 1)$

$\rightsquigarrow$

$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\iddots$
y^3	xy^3	x^2y^3	x^3y^3	x^4y^3	$\cdots$
y^2	xy^2	x^2y^2	x^3y^2	x^4y^2	$\cdots$
y	xy	x^2y	x^3y	x^4y	$\cdots$
1	x	x^2	x^3	x^4	$\cdots$

$\trianglelefteq \mathbb{C}[x, y]$.

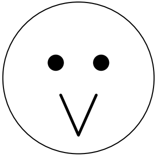
The fiber of Procesi bundle $\mathcal{P}$ at I_μ is a bigraded S_n-module. By Haiman, the Frobenius character is

$$\mathrm{Frob}_{q,t}(\mathcal{P}|_{I_\mu}) = \widetilde{H}_\mu(X; q, t).$$

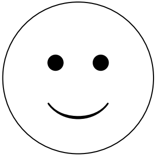
Section 3

Birkoff–Grothendieck Theorem

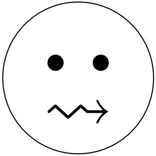
linear algebra over field $\mathbb{C}$



linear algebra over a curve

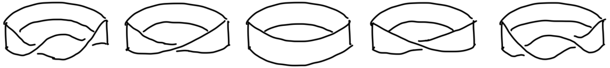


linear algebra over a surface



Birkoff–Grothendieck Theorem

The line bundles (vector bundles of rank 1) over $\mathbb{P}^1$ are parametrized by $\mathbb{Z}$

$\cdots$  $\cdots$

$$\cdots \quad \mathcal{O}_{\mathbb{P}^1}(-2) \quad \mathcal{O}_{\mathbb{P}^1}(-1) \quad \mathcal{O}_{\mathbb{P}^1} \quad \mathcal{O}_{\mathbb{P}^1}(1) \quad \mathcal{O}_{\mathbb{P}^1}(2) \quad \cdots$$

For higher ranks, we have the following decomposition.

Theorem (Birkoff–Grothendieck)

Any vector bundle over $\mathbb{P}^1$ can be decomposed into a direct sum of line bundles.

Compare

over $\mathbb{C}$		over $\mathbb{P}^1$
A **vector space** can be decomposed into **lines** (**vector spaces** of **dimension** 1).	V.S.	A **vector bundle** can be decomposed into **line bundles** (**vector bundles** of **rank** 1).

More generally, let G be a reductive group. We have

over $\mathbb{C}$		over $\mathbb{P}^1$
A G-**representation** can be decomposed into **irreps**.	V.S.	A G-**equivariant vector bundle** can be decomposed into **???**.

Harder–Narasimhan Filtration

In the decomposition, the multiplicity of $\mathcal{O}_{\mathbb{P}^1}(i)$ is unique. But the decomposition is **NOT canonical**.

What is canonical is the subbundle

$$\sum_{j \geq i} \begin{pmatrix} \text{summands} \\ \cong \mathcal{O}_{\mathbb{P}^1}(j) \end{pmatrix} \qquad \text{(direct sum)}.$$

This forms the **Harder–Narasimhan filtration** of the vector bundle. Actually, it is the subsheaf generated by the image

$$H^0\left(\mathbb{P}^1, \mathcal{E} \otimes \mathcal{O}_{\mathbb{P}^1}(-i)\right) \otimes \mathcal{O}_{\mathbb{P}^1}(i) \longrightarrow \mathcal{E}.$$

Equivariant Birkoff–Grothendieck Theorem

Assume

$$\boxed{\begin{array}{c}\mathcal{O}_{\mathbb{P}^1}(1) \text{ can be equipped with} \\ \text{a structure of } G\text{-equivariant line bundle.}\end{array}} \qquad (*)$$

What we have discussed shows the following

Theorem

Under $(*)$, any G-equivariant vector bundle over $\mathbb{P}^1$ can be decomposed into a direct sum of $\mathcal{O}_{\mathbb{P}^1}(i) \otimes V$ for $i \in \mathbb{Z}$ and V an irrep.

- Here we can choose an equivariant splitting since $(-)^G$ preserves surjectivity.
- The condition $(*)$ is true for any torus and GL_2, but not for PGL_2.

Return to Hilbert Schemes

Recall

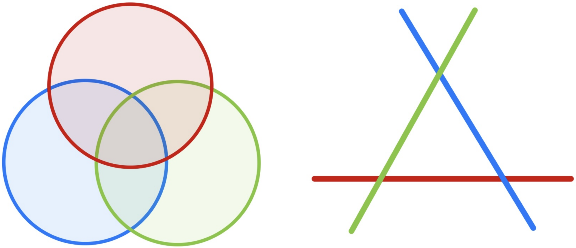
$$\begin{tikzcd}
X_n \arrow[r, equal] \arrow[dd] & \mathbb{C}^2 \times \begin{pmatrix} \text{total space} \\ \text{of } \mathcal{O}_{\mathbb{P}^1}(-1) \end{pmatrix} \arrow[d, "\text{square}"'] & \\
& & \\
H_n \arrow[r, equal] & \mathbb{C}^2 \times \begin{pmatrix} \text{total space} \\ \text{of } \mathcal{O}_{\mathbb{P}^1}(-2) \end{pmatrix} & \mathbb{P}^1 \arrow[l, "\supset"']
\end{tikzcd}$$

When restricting to $\mathbb{P}^1$, the Procesi bundle decomposed into

$$\mathcal{P}|_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1} \otimes \mathsf{tri} \oplus \mathcal{O}_{\mathbb{P}^1}(1) \otimes \mathsf{sgn}.$$

Section 4

The Proof and Further



More Properties

To proceed the proof, we need two properties, more or less following from Haiman's proof.

The Procesi bundle $\mathcal{P}$ is generated by global sections.

$$\mathcal{O}_{H_n\times\mathbb{C}^{2n}} \twoheadrightarrow \mathcal{O}_{X_n}$$

$$\begin{tikzcd}
X_n \arrow[rr] \arrow[dr, "\text{closed}"] \arrow[dd, "\rho"'] & & \mathbb{C}^{2n} \arrow[dd] \\
& H_n \times \mathbb{C}^{2n} \arrow[ur] \arrow[dl, "\text{affine}"] & \\
H_n \arrow[rr, "\sigma"] & & S^n\mathbb{C}^2
\end{tikzcd}$$

We have

$$\mathcal{P} \xrightarrow{\sim} \mathcal{P}^\vee \otimes \mathcal{O}_{H_n}(1) \otimes \mathbf{sgn}.$$

This is the geometric version of the reciprocity identity

$$\widetilde{H}_\mu = T_\mu \cdot \omega \widetilde{H}_\mu \big|_{\substack{q\mapsto q^{-1} \\ t\mapsto t^{-1}}}.$$

Recall $T_\mu = t^{n(\mu)} q^{n(\mu')}$.

The Proof

Assume that μ is obtained by moving a box of λ.

Geometrically, we can connect the two points $I_\lambda, I_\mu \in H_n$ by a T-equivariant curve $\mathbb{P}^1$. Restricting to this $\mathbb{P}^1$, one can show the Procesi bundle splits into

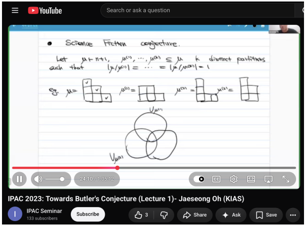
$$\mathcal{P}|_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1} \otimes M \oplus \mathcal{O}_{\mathbb{P}^1}(1) \otimes N$$

for bigraded S_n-representations M, N. Localizing at I_λ and I_μ, we get

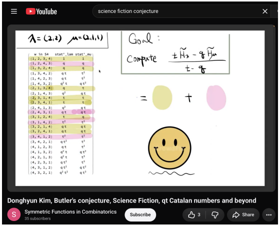
$$\widetilde{H}_\lambda = 1 \cdot \mathrm{Frob}_{q,t}(M) + T_\lambda \cdot \mathrm{Frob}_{q,t}(N),$$
$$\widetilde{H}_\mu = 1 \cdot \mathrm{Frob}_{q,t}(M) + T_\mu \cdot \mathrm{Frob}_{q,t}(N).$$

We are done.

Science Fiction Conjecture



talk by Jaeseong Oh



talk by Donghyun Kim

Conjecture (Part of Science Fiction Conjecture)

Let $\lambda \vdash n+1$ and $\mu(1), \ldots, \mu(k)$ with $|\lambda/\mu(i)| = 1$. Then

$$\sum_{i=1}^{k} \prod_{j \neq i} \frac{T_{\mu(i)}}{T_{\mu(i)} - T_{\mu(j)}} \widetilde{H}_{\mu(i)}(X; q; t) \in \sum \mathbb{N}[q, t] s_{\theta}(X).$$

Cohomological Vanishing Conjecture

Similarly, there is an embedding

$$f : \mathbb{P}^{k-1} \longrightarrow H_n$$

such that the T–fixed points are sent to $I_{\mu(1)}, \ldots, I_{\mu(k)}$. By Atiyah–Bott localization theorem, we can show

Proposition

$$\left(\begin{array}{c} \textit{the expression} \\ \textit{in the conjecture} \end{array} \right) = \sum_{i \geq 0} (-1)^i \, \mathsf{Frob}_{q,t} \left(H^i(\mathbb{P}^1, f^* \mathcal{P}^\vee)^* \right).$$

As a result, the conjecture can be implied by the following vanishing conjecture

Conjecture

$$H^i(\mathbb{P}^1, f^* \mathcal{P}^\vee) = 0, \qquad \text{for any } i > 0.$$

Thanks